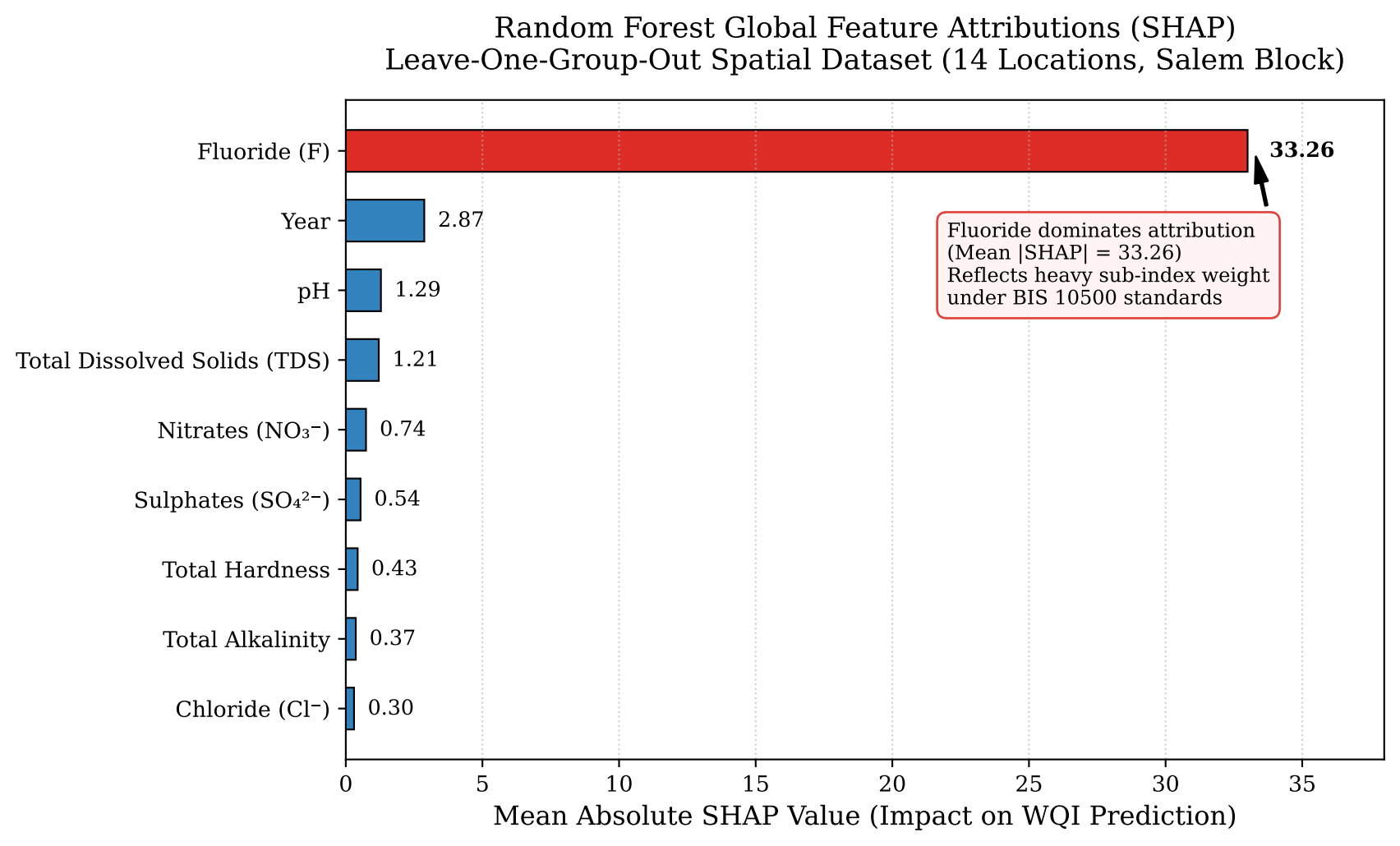
Random Forest Global Feature Attributions (SHAP)
Leave-One-Group-Out Spatial Dataset (14 Locations, Salem Block)
Fluoride (F)
Year
pH
Total Dissolved Solids (TDS)
Nitrates (NO₃⁻)
Sulphates (SO₄²⁻)
Total Hardness
Total Alkalinity
Chloride (Cl⁻)
33.26
2.87
1.29
1.21
0.74
0.54
0.43
0.37
0.30
Fluoride dominates attribution
(Mean |SHAP| = 33.26)
Reflects heavy sub-index weight
under BIS 10500 standards
0
5
10
15
20
25
30
35
Mean Absolute SHAP Value (Impact on WQI Prediction)